

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course : Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTETPE603A - MICROPROCESSORS AND MICROCONTROLLERS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The 8085 microprocessor has how many address lines?
a) 8 b) 16 c) 32 d) 64
- The instruction ***MVI A, 32H*** takes how many bytes?
a) 1 b) 2 c) 3 d) 4
- The default register pair used for memory pointer is:
a) BC b) DE c) HL d) PSW
- Which flag is not available in the 8085 flag register?
a) Sign b) Zero c) Carry d) Overflow
- 5 The ***XCHG*** instruction exchanges data between:
a) HL and DE b) BC and DE c) SP and PC d) HL and BC
- 6 Which instruction does not affect any flag?
a) INR M b) MOV A, B c) DCR A d) ADD C
- 7 The maximum memory addressable by 8085 is:
a) 64 KB b) 32 KB c) 16 KB d) 256 KB
- The first machine cycle of any instruction is always:
a) Read b) Opcode fetch c) Write d) Memory fetch
- Which register in 8051 is used to hold the address of the Stack Top?
a) DPTR b) PC c) SP d) PSW
- How many timers are available in a standard 8051 microcontroller?
a) 1 b) 2 c) 3 d) 4
- 11 The 8051 has how many I/O ports?
a) 2 b) 3 c) 4 d) 5
- 12 What is the function of the ***ANL*** instruction?
a) OR b) AND c) XOR d) Shift

Q2. Solve the following.

- A) Describe architecture of an 8085 microprocessor with neat diagram. 6
- B) Draw pin diagram of 8085 microprocessor and describe the functions of all pins of it. 6
- Q3. Solve the following.
- A) Describe following instructions of 8085 microprocessor with suitable examples of each. 6
- a) LDA 2000H ; b) ANA M
- B) Describe how stack operations are performed in 8085 microprocessor. 6
- Q4. Solve Any Two of the following.
- A) Draw timing diagram for *ADI 55H* and *MOV A, M* instructions of 8085 microprocessor. 6
- B) Find out total time required for execution of following 8085 program. Assume that operating frequency of 8085 is 3 MHz. Also mention total size of following program. 6
- ```

LXI H, 2050H
MOV A, M
INX H
ADD M
INX H
MOV M, A
HLT

```
- What will be the status of S, Z, AC, CY, and P flags after execution of following program. Also mention content of accumulator after execution of following program. 6
- ```

LDA 1000H
XRA A
MOV B, A
INR B
LDA 1001H
ORI 0FFH
ADD B
HLT

```
- Q5. Solve Any Two of the following.
- A) Describe PSW or Flag register of 8085 microprocessor. 6
- B) Describe RAM memory structure of an 8051 microcontroller. 6
- C) Draw pin diagram of an 8051 microcontroller and describe functions of any 4 pins of it. 6
- Q6. Solve Any Two of the following.
- A) Describe function of all bits of TMOD register of an 8051 microcontroller. 6
- B) Assume that an LED is connected to pin P1.1. Write an assembly language program to blink LED continuously with a delay of 10 ms. Assume XTAL = 11.0592 MHz. 6
- C) Interface an 16x2 LCD to 8051 microcontroller and write a program to display message 'ABCD' on it. 6

*** End ***